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Dialogue-Centered Knowledge Architecture: From LLM-Wiki to Persistent Human–AI Semantic Systems

Abstract Draft v1

Large language model (LLM) systems are increasingly evolving beyond retrieval-centered workflows toward persistent semantic architectures. While Retrieval-Augmented Generation (RAG) systems primarily reconstruct semantic organization transiently at query time, emerging approaches such as LLM-Wiki introduce persistent semantic compilation layers that support incremental semantic organization across interactions.

This paper proposes a dialogue-centered knowledge architecture for persistent human–AI semantic systems. In the proposed framework, persistent dialogue functions not merely as an access mechanism for stored knowledge, but as a continuous semantic interaction process that supports semantic continuity, iterative rectification, and long-duration structural stabilization.

The architecture integrates several components, including semantic compilation, persistent dialogue layers, role-distributed semantic functions, operational governance mechanisms, glossary stabilization, and human–AI co-rectification processes. Together, these components support continuously maintainable semantic structures that can be externally projected into papers, figures, glossaries, and explanatory layers.

The paper further examines how governance and stabilization mechanisms may help constrain uncontrolled semantic expansion while preserving long-duration semantic continuity across interactions. Three conceptual figures are provided to illustrate the transition from retrieval-centered systems toward dialogue-centered semantic architectures, their layered internal organization, and their dynamic co-rectification processes.

Rather than presenting a benchmark or implementation-focused system, the paper aims to provide a conceptual architecture for understanding persistent semantic organization in long-duration human–AI collaborative environments.

Table of Contents (Draft)

Abstract

Introduces the transition from retrieval-centered systems toward persistent semantic architectures and proposes a dialogue-centered knowledge architecture for long-duration human–AI semantic systems.

1. Introduction

Discusses the limitations of purely retrieval-centered workflows, the emergence of persistent semantic systems such as LLM-Wiki, and the growing importance of long-duration semantic continuity.

2. From Retrieval to Semantic Compilation

Examines the transition from conventional Retrieval-Augmented Generation (RAG) architectures toward persistent semantic compilation systems that maintain semantic organization across interactions.

3. Persistent Semantic Architectures

Discusses the structural limitations of document-centered knowledge systems and introduces persistent semantic organization as a continuously maintainable semantic layer.

4. Dialogue-Centered Knowledge Architecture

Introduces the proposed architecture in which persistent dialogue functions as a semantic coordination mechanism supporting continuity, stabilization, and long-duration semantic maintenance.

Subsections include:

- Persistent Dialogue

- Phase Structuring and Semantic Organization
- Distributed Semantic Roles
- Human–AI Co-Rectification

5. Human–AI Co-Rectification and Governance

Examines semantic rectification, operational governance, distributed semantic coordination, and long-duration semantic stabilization processes within persistent semantic environments.

6. Risks and Structural Limitations

Discusses structural risks associated with persistent semantic systems, including semantic rigidity, persistent instability, structural propagation of misinterpretation, and governance-related limitations.

7. Discussion

Situates dialogue-centered semantic architectures within the broader evolution of long-duration human–AI semantic systems and discusses persistent semantic continuity, semantic infrastructure, and external semantic compatibility.

8. Conclusion

Summarizes the proposed framework and discusses dialogue-centered semantic architectures as a possible direction in the ongoing evolution of persistent human–AI semantic systems.

Figures

Figure 1.
Transition from Retrieval-Centered Systems to

Dialogue-Centered Semantic Architectures

Figure 2.

Layered Conceptual Architecture for Persistent
Human–AI Semantic Systems

Figure 3.

Human–AI Co-Rectification Loop in Persistent
Semantic Systems